

Exploring Disparate Effects Among Sex and Age in Activity-Based Workplace Implementation

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2025-04-28

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1 Abstract

Ensuring equity for all employees in one's workplace is an important component to consider with any wide scale organizational change. This study uses a pre-test/post-test ANCOVA design to examine the implementation of an Activity Based Work environment, a method of creating flexible work spaces. This study aims to test whether there are disparate effects among sex and age groups in their job satisfaction. Understanding how a company's demographics is impacted by workplace initiatives is important to address when considering whether this strategy is a good fit for the workplace.

2 Introduction

Activity Based Work environments is a method of creating a flexible working environment in which employees can choose their preferred mode of work as it suites their needs. This can be an important tool for work spaces looking to give more agency to their employees while also potentially fostering benefits. It is important to consider that with any large scale change whether or not certain populations of people are disparately affected by the change. For example, a study by (Haapakangas, Sirola, and Ruohomäki 2022) seems to indicate that people who are younger in age tend to utilize the new office space design more than older individuals. This being the case, it is more likely that the effects will not be uniform across all demographics in each setting. This study used a publicly available dataset from (Halldórsson 2024).

3 Method

3.1 Survey Design

Workers were surveyed one month before implementation. Then, they were surveyed three more times at one month, six months, and 12 months after implementation. For the purposes of this study, only the surveys taken before implementation and 12 months after implementation were used for the pre-test/post-test design. The survey questions used a 5 item likert scale. Additional demographic data was collected. It is worth noting that while the entire dataset was cleaned, only job satisfaction and demographic data were used for the purpose of testing this hypothesis in a separate dataset created from that set.

3.2 EDA/Cleaning

Each likert scale score was summed together and divided by the number of questions in it's respective category to get an average score for each construct. Questions were reverse coded as needed. Correlation plots were ran on each set of questions to ensure that each question correlated with each other to verify the validity of the hypothetical construct.

Table 1: Data Dictionary

Variable	Description	Type	Range
gender	Participant gender	Factor	0-1
age	Age in years	Numeric	18-65
t1	Job satisfaction before implementation	Numeric	1-5
t4	Job satisfaction after 12 months	Numeric	1-5

A table containing the variable name, a description of that variable, it's file class, and range of values.

3.3 Hypotheses

We hypothesize first that there will be an effect on job satisfaction in regards to age after 12 months into the implementation of ABW. We also hypothesize that there will be an effect on sex after 12 months into the implementation of ABW. The IVs were age, sex, and job satisfaction 1 month before implementation (t1). Both hypotheses are tested with one model. The Analysis of Covariance (ANCOVA) model was deemed appropriate to test these hypotheses.

A table containing the variable name, it's mean, standard deviation, minimum score, and maximum score.

Table 2: Data Excerpt

variable	mean	sd	min	max
age	53.821429	9.0022778	33	67
t1	4.030303	0.8191020	2	5
t4	4.171717	0.7101507	3	5

3.4 Missingness and Imputation

The nature of the original data set had 56%-76% missing data, with some participants missing entire testing sessions. In order to get as much usefulness out of the available data, the Available Items imputation method was employed. The method worked by using mean imputation only on the items from the same construct and from the same timepoint to preserve the non-independant nature of the data. If a participant was missing one timepoint, data from the other 3 timepoints were used for mean imputation within the items of that construct. Participants who missed more than one time point were cut from the study. While a custom function was used to achieve results, adherence to these rules should yield the same outcome.

Table 3: Cleaned Data

studyid	age	gender	t1	t2	t3	t4
A10	33	0	4.0	5.000000	4.666667	5.0
A11	56	0	3.5	4.000000	3.000000	3.5
A12	67	0	2.5	2.666667	2.500000	3.0
A14	59	0	3.5	3.166667	3.000000	3.0
A15	NA	0	5.0	4.666667	5.000000	4.0

A table of a sample of the dataset displaying the participant's id, age, gender, and timepoints 1 - 4.

4 Results

4.1 Hypothesis 1:

We found that the first hypothesis was significant and indicated a negative relationship on job satisfaction after 12 months in ABW implementation ($b = -0.022$, $t = -2.27$, $p = < 0.05$). For every additional unit increase in age, a participant's job satisfaction after treatment would decrease by .022 points in magnitude on a 1-5 scale.

4.2 Hypothesis 2:

We found that the second hypothesis was not significant indicating that gender does not significantly affect job satisfaction after 12 months in ABW implementation ($b = .27$, $T = 1.32$, $p = > .05$).

4.3 Findings Figure and Equation:

$$Y = 2.416 + 0.27 \cdot \text{gender} + -0.022 \cdot \text{age} + 0.717 \cdot \text{t1} + \epsilon$$

Table 4: ANCOVA Model Results

term	estimate	std.error	statistic	p.value
(Intercept)	2.4164031	0.6087380	3.969529	0.0005691
gender1	0.2700989	0.2050574	1.317187	0.2002142
age	-0.0215891	0.0095233	-2.266980	0.0326829
t1	0.7173710	0.0991388	7.236027	0.0000002

5 Discussion

The results of the study show that in this sample, age could be a marker for assessing how an individual's job satisfaction will improve or in some cases diminish with the treatment. The older the worker is the less satisfied they will be with their job, after one year in the new ABW environment. Conversely, younger employees saw a much greater improvement in their workplace satisfaction. This effect is also reflected by the notion that younger employees tend to be more resourceful in utilizing ABWs (Haapakangas, Sirola, and Ruohomäki 2022).

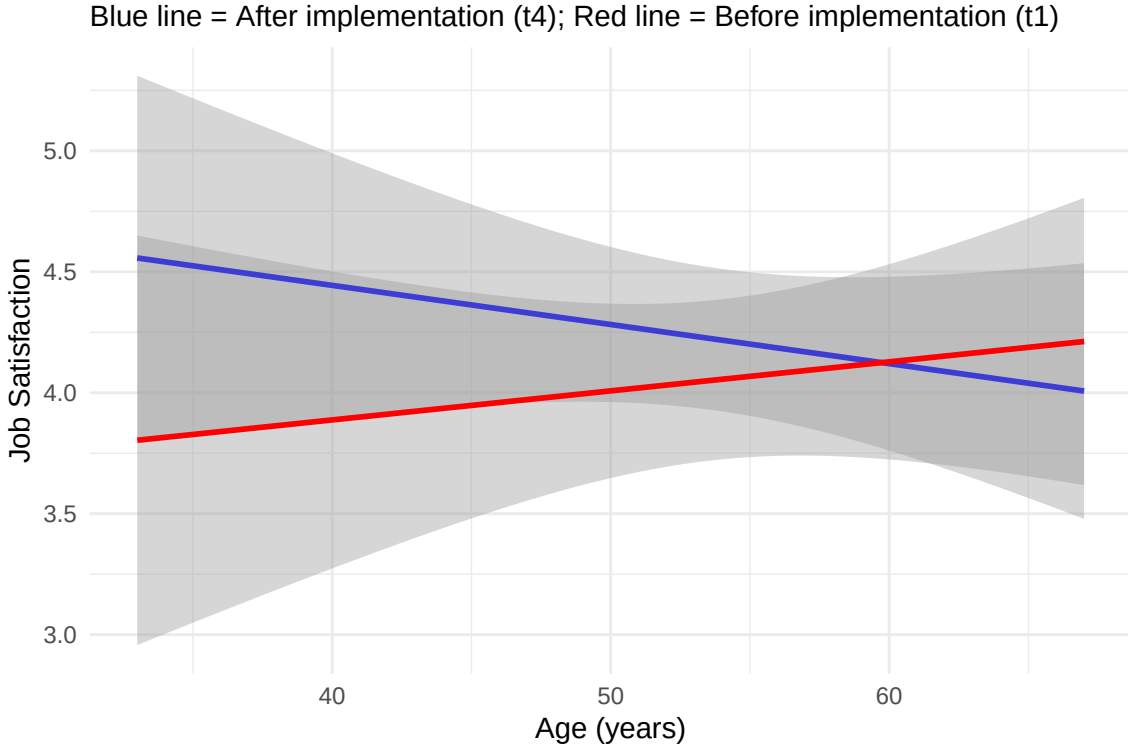


Figure 1: A graph showing the trend in job satisfaction experiences by age. Job satisfaction scores are on the y axis and age in years is on the x axis.

It is important to consider what could potentially lead to this effect for older employees. A study by (Van Vianen et al. 2022) seems to suggest that older employees can experience more maladaptability to workplace changes, and this often disrupts facets of their home life. Employees that experience more workplace change seem to report more sleep disruptions at home (Van Vianen et al. 2022). This impact from work life to home life may potentially explain the growing dissatisfaction in older employees. Additionally, it can be difficult for older workers to unlearn parts of their old routine (Van Vianen et al. 2022). This evidence shows how large scale changes have effects on their employees outside of the workplace. Despite these findings it is important to consider that older employees tend to hold many more emotional resources gained

from their longer professional experience (Kim and Kang 2016). Older employees are also more engaged in their work than younger employees (Kim and Kang 2016). Combining the results of this study and these other perspectives on the aging workforce, shows the importance for HR practitioners to exercise caution in enabling new workplace initiatives. Failure to account for these individual differences in your workplace environment may result in systemic ageism in the workplace. Potential solutions to mitigate the drawbacks for the older population may exist but more research on this topic is required.

The effect of sex on implementation efficacy not being significant is beneficial to the practice of ABW. This shows that across sex, there is no found evidence of disparate effects. It's important to note that experiences across sex can heavily vary culturally and that it is up to the HR practitioner to be aware of when this may be an issue.

6 Limitations and the Next Step

The sample size of this study is small. After handling missingness, the study has 33 observations. Additionally, this study is best to be considered as preliminary as the current version of this study does not fully utilize the dataset's timepoints between t1 and t4. To best leverage the full content of the dataset, A Linear Mixed Effects Model would be could be appropriate. Some of the missing data could potentially be explained by other variables in the data set such as turnover intentions. This would be an important variable to explore in the future to potentially explore if the data is missing at random. Additionally, its important to consider the cultural differences in workplaces globally as a limitation across generalizability to workplaces in other nations.

References:

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